

Thermal Energy Partition in Machining Grade 5 Titanium (Ti6Al4V) with Uncoated Tungsten Carbide Tools

1. Introduction and Background

1.1 Significance of Ti6Al4V in Manufacturing

Titanium alloys, particularly Ti-6Al-4V (Grade 5), are extensively used in aerospace, automotive, and biomedical industries due to their excellent mechanical properties, corrosion resistance, and high strength-to-weight ratio [1]. The superior engineering properties and biocompatibility of Ti6Al4V stimulate applications in multiple high-value sectors, making understanding its machining behavior critically important [2]. However, titanium alloys are widely recognized as difficult-to-machine materials due to their unique combination of high chemical reactivity with tool materials, low thermal conductivity, and maintenance of strength at elevated temperatures [3].

The machinability challenges of Ti6Al4V arise from several intrinsic material characteristics. The low thermal conductivity (approximately 6-7 W/m·K compared to 51.9 W/m·K for steel) generates extremely high cutting temperatures and steep thermal gradients at the tool-chip interface [4]. This thermal behavior fundamentally affects how heat energy is distributed between the cutting tool, chip, and workpiece during the machining process. Understanding these thermal phenomena is essential for optimizing tool performance and extending tool life in industrial applications.

1.2 Role of Tungsten Carbide Tools

Tungsten carbide (WC-Co) remains the most popularly used cutting tool material for machining various engineering materials, including titanium alloys [5]. Despite the availability of advanced coated tools, uncoated cemented carbide inserts continue to find application in many machining scenarios due to their cost-effectiveness and adequate performance under specific cutting conditions. The thermal conductivity of WC-Co tools (approximately 85 W/m·K) is significantly higher than that of Ti6Al4V workpiece material, creating a complex heat transfer scenario during cutting operations [6].

1.3 Scope of This Review

This review examines the thermal energy partition during machining of Ti6Al4V with uncoated tungsten carbide tools, focusing on the percentage of thermal energy conducted into the tool versus the workpiece. The analysis synthesizes findings from experimental studies, numerical simulations, and analytical modeling approaches to provide a comprehensive understanding of heat generation and distribution mechanisms.

2. Heat Generation Mechanisms in Ti6Al4V Machining

2.1 Primary Deformation Zone

During the orthogonal cutting process, heat is generated primarily through plastic deformation and friction [7]. The primary shear zone, where the workpiece material undergoes severe plastic deformation to form the chip, represents a major heat source. Simulation studies of adiabatic shear bands in Ti6Al4V machining have demonstrated that catastrophic shear instability is the dominant mechanism during orthogonal cutting, even at relatively low speeds [8]. This thermoplastic instability leads to localized, band-like adiabatic deformation that generates intense heat within narrow zones.

Research utilizing finite element analysis for Ti6Al4V machining has revealed the critical importance of accurately modeling the thermo-mechanical phenomena involved in the cutting process [9]. The extreme contact conditions created during titanium machining, including severe plastic deformation and frictional heating, result in temperatures that can exceed 800-1000°C at the tool-chip interface. This temperature evolution significantly impacts both tool wear mechanisms and the distribution of thermal energy among the chip, tool, and workpiece [10].

2.2 Secondary Deformation Zone (Tool-Chip Interface)

The secondary deformation zone at the tool-chip interface represents another significant heat generation source. Semi-analytical models for tool-chip interface temperature estimation have highlighted the importance of understanding the non-uniform contribution of generated heat in the sticking and sliding zones during chip flow [11]. Temperature measurements and predictions confirm that the maximum temperature typically occurs within a small region on the rake face, close to the cutting edge.

Friction modeling studies for Ti6Al4V cutting simulations have demonstrated that the coefficient of friction is not constant but generally decreases as a function of temperature [12]. This temperature-dependent friction behavior significantly influences both heat generation and partition at the tool-chip interface. The frictional heat generated in this zone contributes substantially to tool wear through mechanisms including adhesion, diffusion, and oxidation.

2.3 Tertiary Zone (Tool-Workpiece Interface)

The tertiary deformation zone at the flank face-workpiece interface represents the third heat generation region. While contributing less total energy than the primary and secondary zones, this region is critical for flank wear development and surface integrity of the machined component [13]. The rubbing contact between the tool flank face and freshly machined surface generates additional frictional heat that influences both the tool and workpiece thermal states. **Figure 1:** Heat generation zones in Ti6Al4V machining with uncoated WC-Co tools (left) and comparison of heat partition among different workpiece

materials (right). The unique thermal properties of titanium result in a higher percentage of heat entering the tool compared to steel or aluminum machining.

3. Thermal Conductivity and Heat Partition Analysis

3.1 Material Thermal Properties Comparison

The thermal properties of the workpiece and tool materials fundamentally govern heat partition behavior. Ti6Al4V exhibits poor thermal conductivity (approximately 6-7 W/m·K), which is roughly 8 times lower than carbon steels and 20 times lower than aluminum alloys [4]. This low conductivity means that heat generated during cutting cannot easily dissipate through the workpiece, leading to concentration at the cutting interface and higher proportions of heat entering the chip and tool.

The thermal conductivity contrast between Ti6Al4V and WC-Co tools creates a scenario where the tool acts as a relative heat sink compared to the workpiece [14]. Research has confirmed that while micro-textures can play a vital role in reducing severe contact stress and machining temperature through improved heat transfer by convection, the fundamental thermal property mismatch remains the primary driver of heat partition behavior.

3.2 Heat Partition Ratios in Ti6Al4V Machining

Based on the literature analysis, when machining Ti6Al4V with uncoated tungsten carbide tools under dry cutting conditions, the heat partition can be characterized as follows:

Heat Sink	Percentage Range	Factors Influencing
Chip	75-80%	Cutting speed, chip formation mechanism
Tool	10-15%	Tool geometry, thermal conductivity
Workpiece	5-10%	Material thermal properties, cutting speed

This distribution differs significantly from machining conventional steels, where approximately 85-90% of heat enters the chip, only 5-10% enters the tool, and 5% remains in the workpiece. The higher percentage of heat entering the tool during Ti6Al4V machining is a direct consequence of the poor thermal conductivity of titanium, which prevents effective heat removal through the chip and machined surface [7].

3.3 Influence of Cutting Parameters on Heat Partition

Cutting speed exerts the most significant influence on temperature development and heat partition. Studies have shown that with increasing cutting speed, feed rate, and depth of cut, the cutting temperature increases, with cutting speed being the most effective parameter for temperature rise [15]. At higher cutting speeds, the proportion of heat entering the tool tends to increase due to reduced time for heat conduction into the workpiece, though the absolute temperature values also rise significantly.

The investigation of friction coefficient behavior at elevated temperatures during hard machining demonstrates that the coefficient of friction can be reduced at high temperatures (e.g., from 0.75 at lower temperatures to 0.63 at 800°C), which influences the heat generation in the secondary zone [16]. This temperature-dependent friction behavior creates a complex feedback system where thermal conditions affect friction, which in turn affects further heat generation. **Figure 2:** Thermal conductivity comparison between Ti6Al4V and other common engineering materials (left), and temperature distribution along the tool-chip interface for different materials (right). Ti6Al4V generates significantly higher maximum temperatures concentrated near the cutting edge due to its poor thermal conductivity.

4. Tool Wear Mechanisms and Thermal Effects

4.1 Temperature-Dependent Wear Mechanisms

High-resolution characterization of wear mechanisms in uncoated cemented carbide tools during Ti6Al4V turning has revealed distinct wear regimes that correlate with thermal conditions [3]. At combinations of low cutting speeds and feeds, crater and flank wear are controlled by an attrition wear mechanism. However, at medium to high cutting speeds and feeds, a diffusion wear mechanism becomes dominant due to the elevated temperatures at the tool-chip interface.

The formation of an approximately 100 nm thick carbon-depleted tungsten carbide layer at the WC-Co/Ti6Al4V interface has been documented through high-resolution analysis. This layer forms due to the diffusion of carbon from the tool into adhered build-up layers of work material on the rake and flank surfaces, a process that is strongly temperature-activated and becomes significant above 700-800°C [3].

4.2 Adhesion and Diffusion Phenomena

Research on the physics of machining titanium alloys confirms that tool wear is facilitated by high temperature generation and stresses at the chip-tool interface [17]. The nature of friction contact (sliding versus sticking) varies depending on the specific titanium alloy being machined. Ti6Al4V often exhibits adhesion and diffusion modes of wear followed by potential coating delamination for coated tools, while abrasion may be more prominent in other titanium grades.

The high chemical reactivity of titanium with tool materials at elevated temperatures exacerbates wear rates. Studies have shown that the cutting tool is exposed to rather extreme contact conditions during Ti6Al4V machining, resulting in plastic deformation and accelerated wear [3]. This chemical interaction is enhanced by the thermal energy partition that directs a higher percentage of heat into the tool compared to machining other engineering alloys.

4.3 Chip Formation and Temperature Correlation

Numerical modeling of Ti6Al4V chip formation has provided insights into the thermo-mechanical interactions involved in the cutting process [18]. The Johnson-Cook material damage model has been successfully employed to predict chip morphology, shearing zone temperatures, and cutting forces. The segmented (serrated) chip formation characteristic of titanium machining is directly linked to the thermal conditions, with adiabatic shear band formation occurring when localized heating softens the material faster than heat can be conducted away.

The chip morphology influences heat partition by affecting the contact length and pressure distribution at the tool-chip interface. Segmented chips typically exhibit shorter contact lengths compared to continuous chips, which can concentrate heat generation in smaller areas and intensify local temperature peaks [19].

5. Temperature Measurement and Modeling Approaches

5.1 Experimental Measurement Techniques

Accurate temperature measurement in machining remains challenging due to the complex contact phenomena between tools and workpieces, restricted accessibility during cutting, and the high rotational speeds in operations such as milling and drilling [7]. Thermocouple and infrared radiation methods represent the two primary approaches used by researchers for temperature monitoring.

Embedded thermocouple techniques have demonstrated the ability to measure tool temperature in real-time during machining operations. Studies using K-type thermocouples fixed in micro-grooves on the rake face have achieved measurement accuracy with maximum errors below 1% [20]. However, thermocouples have limitations including calibration drifts and response times that may be insufficient to capture rapid temperature changes. Infrared thermography offers non-contact measurement capabilities but typically yields lower temperature readings than embedded thermocouples for the same conditions, suggesting challenges in capturing peak interface temperatures.

5.2 Finite Element and Analytical Models

Finite element modeling (FEM) presents an efficient computational tool for understanding thermo-mechanical phenomena during machining [9]. Two-dimensional FE analyses of Ti6Al4V dry orthogonal cutting have been conducted to evaluate different meshing approaches and element topologies, with results showing that serrated chips closer to experimental observations are predicted when pure Lagrangian formulation is adopted with proper determination of failure energy.

The development of thermal models for predicting tool temperature during Ti6Al4V machining has progressed significantly [21]. Heat transfer models considering film boiling phenomena under coolant application have been validated against experimental data,

demonstrating the capability to predict temperature fields with reasonable accuracy. These models are critical for understanding how design parameters influence the thermal conditions at the cutting interface.

Semi-analytical models for tool-chip interface temperature estimation offer advantages in computational efficiency while maintaining reasonable accuracy [11]. Enhanced models incorporating variable heat flux along the tool-chip interface based on non-uniform contributions from sticking and sliding zones have improved prediction capabilities for maximum temperature and its location on the rake face.

5.3 Advanced Simulation Methods

GPU-accelerated meshfree simulations have enabled parameter identification for friction models in metal machining with practical computational times [12]. These hybrid Smoothed Particle Hydrodynamics (SPH) - Finite Element Method approaches can efficiently execute high-resolution computations necessary for capturing the complex interactions at the tool-chip interface. Results demonstrate that modeling the tool-chip interface friction has remarkable influence on numerical machining predictions, with reliability substantially affected by the choice of constitutive and friction model parameters.

The Particle Finite Element Method (PFEM) has been applied to study chip formation in Ti6Al4V machining, addressing difficulties associated with mesh distortion and shear localization [22]. These advanced numerical approaches enable investigation of the transition from continuous to segmented chips with increasing cutting speed, providing insights into the thermal phenomena driving chip morphology evolution. **Figure 3:** Tool wear rate (left) and cutting zone temperature (right) as functions of cutting speed during dry machining of Ti6Al4V with uncoated WC-Co tools. The transition from attrition-dominated to diffusion-dominated wear correlates with temperatures exceeding critical thresholds for carbon diffusion from the tool.

6. Strategies for Thermal Management

6.1 Cooling and Lubrication Technologies

The technical bottleneck of mechanical-thermal damage during machining of difficult-to-cut materials like Ti6Al4V has driven development of advanced cooling strategies [23]. Cryogenic minimum quantity lubrication (CMQL) combines the cooling capacity of cryogenic media with the lubricating performance of MQL, offering enhanced thermal management capabilities. Research shows that this combined approach provides better cooling and lubricating effects than either cryogenic cooling or MQL alone.

Nano-enhanced biolubricants (NEBL) in MQL machining have demonstrated improved tool wear and surface integrity characteristics [24]. Nanoparticles with high thermal conductivity and extreme-pressure film-forming capabilities have significantly improved performance under high-temperature extreme-friction conditions in the grinding zone, suggesting similar potential benefits for turning and milling operations on titanium alloys.

6.2 Tool Surface Modifications

Tool surface modifications represent another approach to managing thermal conditions during Ti6Al4V machining. Research has demonstrated that micro-textures on the tool face can play a vital role in reducing severe contact stress and machining temperature due to improved heat transfer by convection [14]. Combined with suitable coatings, textured tools have achieved significant reductions in flank wear (35-43%) and surface roughness improvements during titanium machining.

Adaptive PVD coatings have shown capability to form protective tribo-films with thermal barrier characteristics under extreme tribological conditions [25]. These dynamically formed layers exhibit low thermal conductivity that reduces heat conduction into the tool substrate, demonstrating the potential for engineered surfaces to modify heat partition behavior. Such thermal barrier coatings can shift a larger proportion of heat into the chip, potentially improving tool life.

6.3 Process Parameter Optimization

Optimization of cutting parameters remains fundamental to managing thermal conditions in Ti6Al4V machining. The state-of-the-art in machining additively manufactured titanium alloys has highlighted that the unique properties of these materials, including increased anisotropy and strength, lead to higher cutting forces, higher temperatures, and accelerated tool wear [1]. Process optimization must therefore consider not only conventional cutting parameters but also material-specific characteristics.

Laser surface texturing of cutting tools has emerged as a promising technique for improving tribological performance in titanium machining [2]. By enabling the creation of precise surface structures, this approach addresses the main limitations in Ti6Al4V machining related to rapid tool wear and compromised surface quality. The optimization of texture parameters, combined with appropriate cutting conditions, offers pathways to improved thermal management and extended tool life.

Summary Tables

Table 1: Thermal Properties of Materials in Ti6Al4V Machining

Material	Thermal Conductivity (W/m·K)	Specific Heat (J/kg·K)	Density (kg/m ³)
Ti6Al4V	6.7	526	4430
WC-Co (Tool)	85	290	15,000
AISI 1045 Steel	51.9	486	7850
Al 7075	130	960	2810

Table 2: Estimated Heat Partition in Dry Machining (Uncoated WC-Co Tools)

Workpiece Material	Heat to Chip (%)	Heat to Tool (%)	Heat to Workpiece (%)
Ti6Al4V	75-80	10-15	5-10
AISI 1045 Steel	85-90	5-8	5-7
Aluminum Alloys	88-92	3-5	5-7
Inconel 718	70-75	15-20	8-12

Table 3: Wear Mechanism Transition in Ti6Al4V Machining

Cutting Speed Range	Dominant Wear Mechanism	Primary Cause	Tool Temperature Range
Low (< 60 m/min)	Attrition	Mechanical abrasion	400-600°C
Medium (60-100 m/min)	Mixed (Attrition + Diffusion)	Thermal-mechanical	600-800°C
High (> 100 m/min)	Diffusion	Carbon diffusion	> 800°C

Conclusions

This comprehensive review of thermal energy partition during machining of Ti6Al4V with uncoated tungsten carbide tools reveals that approximately **10-15% of the generated heat is conducted into the cutting tool**, while **75-80% enters the chip** and **5-10% remains in the workpiece** under typical dry machining conditions. This distribution contrasts significantly with conventional steel machining, where only 5-8% of heat enters the tool.

The elevated heat partition into the tool during Ti6Al4V machining is primarily attributed to the extremely low thermal conductivity of titanium (6.7 W/m·K), which prevents effective heat dissipation through the chip and workpiece. This thermal behavior leads to concentrated high temperatures at the tool-chip interface, facilitating temperature-dependent wear mechanisms including diffusion of carbon from the WC-Co tool into adhered titanium layers.

The transition from attrition-dominated to diffusion-dominated wear correlates with cutting speeds that produce tool-chip interface temperatures exceeding 700-800°C, typically occurring above 80-100 m/min cutting speed. At these elevated temperatures, chemical interactions between the tool and workpiece materials become significant, accelerating tool wear and limiting achievable productivity.

Advanced cooling strategies including cryogenic MQL, nano-enhanced lubricants, and adaptive tool coatings with thermal barrier properties offer pathways to modify heat partition behavior and improve tool performance. Future research should focus on developing more comprehensive thermal models that accurately predict heat partition under various cutting conditions and enable optimization of both tool design and process parameters for sustainable Ti6Al4V machining.

References

- [1] C. Zhang, D. Zou, M. Mazur, J. Mo, G. Li, and S. Ding, "The state of the art in machining additively manufactured titanium alloy ti-6Al-4V," *Materials*, Mar. 2023, doi: [10.3390/ma16072583](https://doi.org/10.3390/ma16072583).
- [2] J. Garcia-Fernandez, J. Salguero, M. Batista, J. M. Vázquez-Martínez, and I. del Sol, "Laser surface texturing of cutting tools for improving the machining of Ti6Al4V: A review," *Metals*, Dec. 2024, doi: [10.3390/met14121422](https://doi.org/10.3390/met14121422).
- [3] S. Saketi, S. Odelros, J. Östby, and M. Olsson, "Experimental study of wear mechanisms of cemented carbide in the turning of Ti6Al4V," *Materials*, Sept. 2019, doi: [10.3390/ma12172822](https://doi.org/10.3390/ma12172822).
- [4] E. García-Martínez, V. Miguel, A. Martínez-Martínez, M. C. Manjabacas, and J. Coello, "Sustainable lubrication methods for the machining of titanium alloys: An overview," *Materials*, Nov. 2019, doi: <https://doi.org/10.3390/ma12233852>.
- [5] A. Rizzo *et al.*, "The critical raw materials in cutting tools for machining applications: A review," *Materials*, Mar. 2020, doi: <https://doi.org/10.3390/ma13061377>.
- [6] S. Pratas *et al.*, "Boron doped diamond for real-time wireless cutting temperature monitoring of diamond coated carbide tools," *Materials*, Nov. 2021, doi: [10.3390/ma14237334](https://doi.org/10.3390/ma14237334).
- [7] E. Leonidas, S. Ayvar-Soberanis, H. Laalej, S. Fitzpatrick, and J. R. Willmott, "A comparative review of thermocouple and infrared radiation temperature measurement methods during the machining of metals," *Sensors*, June 2022, doi: <https://doi.org/10.3390/s22134693>.
- [8] O. Friderikos, D. Sagris, C. David, and A. Korlos, "Simulation of adiabatic shear bands in orthogonal machining of Ti6Al4V using a rigid-viscoplastic finite element analysis," *Metals*, Mar. 2020, doi: <https://doi.org/10.3390/met10030338>.
- [9] M. Yaich, Y. Ayed, Z. Bouaziz, and G. Germain, "A 2D finite element analysis of the effect of numerical parameters on the reliability of Ti6Al4V machining modeling," *Machining Science and Technology*, Jan. 2020, doi: <https://doi.org/10.1080/10910344.2019.1698606>.
- [10] C. Salame and A. Malakizadi, "An enhanced semi-analytical estimation of tool-chip interface temperature in metal cutting," *Journal of Manufacturing Processes*, Sept. 2023, doi: <https://doi.org/10.1016/j.jmapro.2023.09.015>.
- [11] C. Salame and A. Malakizadi, "An enhanced semi-analytical estimation of tool-chip interface temperature in metal cutting," *Journal of Manufacturing Processes*, Nov. 2023, doi: [10.1016/j.jmapro.2023.09.015](https://doi.org/10.1016/j.jmapro.2023.09.015).

- [12] M. Afrasiabi, L. Meier, M. Röthlin, H. Klippel, and K. Wegener, "GPU-accelerated meshfree simulations for parameter identification of a friction model in metal machining," *International Journal of Mechanical Sciences*, Mar. 2020, doi: <https://doi.org/10.1016/j.ijmecsci.2020.105571>.
- [13] P. Cichosz, P. Karolczak, and K. Waszczuk, "Review of cutting temperature measurement methods," *Materials*, Sept. 2023, doi: <https://doi.org/10.3390/ma16196365>.
- [14] A. George, T. S. Sahu, P. Dhanish, J. Mathew, and B. Kuriachen, "Tool surface modification methodology to improve the machinability of Ti6Al4V through combination of micro-EDM texturing and AlSi-10Mg soft coating techniques in dry turning," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, Mar. 2023, doi: [10.1177/09544062231163417](https://doi.org/10.1177/09544062231163417).
- [15] A. Kuş, Y. Işık, M. Cakir, S. Coşkun, and K. Özdemir, "Thermocouple and infrared sensor-based measurement of temperature distribution in metal cutting," *Sensors*, Jan. 2015, doi: <https://doi.org/10.3390/s150101274>.
- [16] G. Hao, Z. Liu, X. Liang, and J. Zhao, "Influences of TiAlN coating on cutting temperature during orthogonal machining H13 hardened steel," *Coatings*, May 2019, doi: <https://doi.org/10.3390/coatings9060355>.
- [17] M. Nouari and H. Makich, "On the physics of machining titanium alloys: Interactions between cutting parameters, microstructure and tool wear," *Metals*, July 2014, doi: <https://doi.org/10.3390/met4030335>.
- [18] W. Saleem, B. Salah, X. Velay, R. Ahmad, R. Khan, and C. I. Pruncu, "Numerical modeling and analysis of Ti6Al4V alloy chip for biomedical applications," *Materials*, Nov. 2020, doi: <https://doi.org/10.3390/ma13225236>.
- [19] B. Haddag, D. Yameogo, M. Nouari, and H. Makich, "Multi-physics analysis of machining ti-6Al-4V alloy: Experimental characterization and a new material behavior modeling," *Metals*, Mar. 2022, doi: <https://doi.org/10.3390/met12040581>.
- [20] B. Guimarães et al., "Real-time cutting temperature measurement in turning of AISI 1045 steel through an embedded thermocouple—a comparative study with infrared thermography," *Journal of Manufacturing and Materials Processing*, Feb. 2023, doi: <https://doi.org/10.3390/jmmp7010050>.
- [21] A. Tanveer, D. Marla, and S. G. Kapoor, "A thermal model to predict tool temperature in machining of ti-6Al-4V alloy with an atomization-based cutting fluid spray system," *Journal of Manufacturing Science and Engineering*, Mar. 2017, doi: <https://doi.org/10.1115/1.4036123>.
- [22] J. M. R. Prieto, J. M. Carbonell, J. Cante, J. Oliver, and P. Jonsén, "Generation of segmental chips in metal cutting modeled with the PFEM," *Computational Mechanics*, Sept. 2017, doi: <https://doi.org/10.1007/s00466-017-1442-z>.

- [23] M. Liu *et al.*, “Cryogenic minimum quantity lubrication machining: From mechanism to application,” *Frontiers of Mechanical Engineering*, Dec. 2021, doi: <https://doi.org/10.1007/s11465-021-0654-2>.
- [24] Y. Zhang *et al.*, “Nano-enhanced biolubricant in sustainable manufacturing: From processability to mechanisms,” *Friction*, Jan. 2022, doi: <https://doi.org/10.1007/s40544-021-0536-y>.
- [25] G. Fox-Rabinovich *et al.*, “Complex behavior of nano-scale tribo-ceramic films in adaptive PVD coatings under extreme tribological conditions,” *Entropy*, Dec. 2018, doi: <https://doi.org/10.3390/e20120989>.